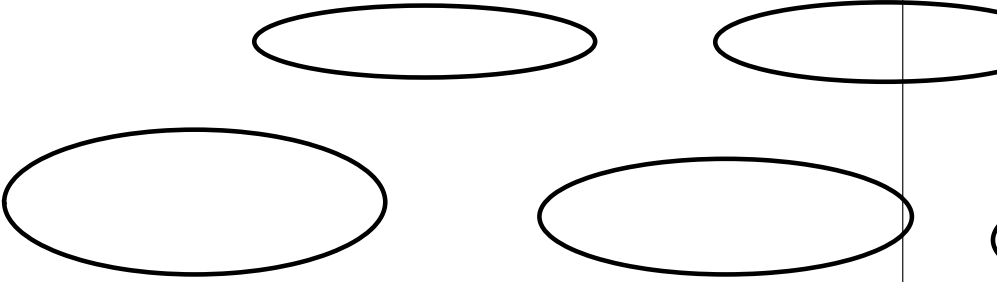
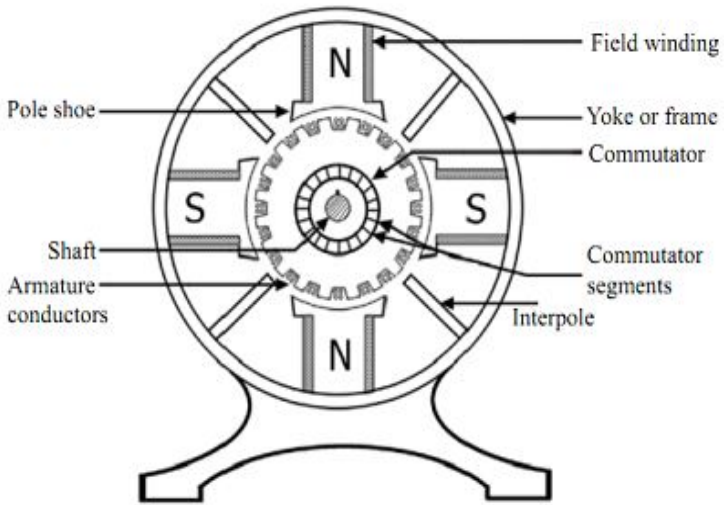


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
FOURTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2022		
Course Code: EET202		
Course Name: DC MACHINES & TRANSFORMERS		
PART A		
		Answer all questions. Each question carries 3 marks
Q1	Why equalizer rings are not used in wave windings?	
Ans	In lap winding, the number of parallel paths is equal to the total number of poles and is used in low voltage and high current dc machines. Under ideal conditions, the flux from all the poles is exactly same, hence the induced emf and current carried by each path will be equal. But under practical operating condition there is variations in air-gap length or difference in magnetic properties of poles material which causes a slight unbalance of induced emf in the various parallel paths. As a result, conductor under stronger poles generates greater emf and carries larger current than remaining conductors. It means that some brushes are overloaded and carry more current than normal current while the remaining brushes carry less current due to under loaded. The current doesn't reverse completely in a given commutation time period due to overloaded brush and that will cause sparking between brushes and commutator surface. The equalizer rings are used to overcome such problem. But in case of wave winding, there is only 2 parallel paths for any number of poles. Thus both the parallel paths are distributed under all the poles. Thus there will not be the issue of unequal current distribution and no need of equalizer rings.	
Q2	Compare lap and wave windings in DC machines	
Ans	LAP WINDING - Finishing end of the first coil can be connected to the starting end of the next coil which starts from the same pole where the first coil started. - Armature windings are divided into	WAVE WINDING Finishing end of the first coil can be connected to the starting end of the next coil which starts from the next similar pole where the first coil started

		several parallel paths • No. of parallel paths = no. of poles • Used for machines having large current & low voltage • No. of brushes required is equal to no. of poles • Equalizer rings may be required • No need of dummy coils • Used in Medium(50-500kW) and high power(>500kW) applications • Commonly used for Current > 400A	• Only two parallel paths independent of number of poles • Used for machines with low current and high voltage • Number of brushes required is always equal to two • No need of equalizer rings • Dummy coils may be required • Used for low power applications • Commonly used for currents < 400A
	Q3	What are the necessary conditions for voltage build up in a DC shunt generator?	
	Ans	• Presence of residual flux • Field connections should be in such a way that field flux should aid the residual flux • Field resistance should be less than critical field resistance • Speed should be greater than the critical speed	
	Q4	What are the advantages and disadvantages of separately excited DC machines?	
	Ans	Advantages • It can operate in stable condition with any field excitation and gives wide range of output voltage while acting as generator • It is good for variable speed control. The field can be connected to a constant voltage source to ensure full torque is available at all speeds. The armature can be connected to a variable voltage DC source to achieve speed control. • It makes the motor easier to reverse. By reversing the polarity of the field supply, the motor will run in reverse. Disadvantages • it is very expensive of providing a separate excitation source. • More terminals in the terminal box	
	Q5	Why the speed of a DC shunt motor is practically constant?	
	Ans	In case of DC shunt motor, flux is practically constant as the field is parallel to the armature. If the load is increased, then the speed of the motor will remain almost	

		constant because the field current remains almost constant. So that it is also known as constant speed motor.
Q6		Explain different losses in a DC motor
		
Q7		Derive the emf equation of a single-phase transformer
		When a sinusoidal voltage of frequency 'f' is applied across the primary windings, it produces a sinusoidal flux of same frequency. Let, the flux at any instant can be given by $\phi = \phi_m \sin \omega t$ According to Faraday's Law, Induced EMF, $e = -N \frac{d\phi}{dt}$ Thus, induced EMF in Primary winding $ \begin{aligned} e_1 &= -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt}(\phi_m \sin \omega t) \\ &= -N_1 \phi_m \omega \cos \omega t \\ &= -N_1 \phi_m \omega \sin(90 - \omega t) \\ &= N_1 \phi_m 2\pi f \sin(\omega t - 90) \end{aligned} $

	So, RMS value of voltage induced in primary winding $E_1 = \frac{N_1 \phi_m 2\pi f}{\sqrt{2}}$ $E_1 = 4.44 N_1 \phi_m f$ Similarly, RMS value of voltage induced in secondary winding $E_2 = 4.44 N_2 \phi_m f$ N_1 : Number of primary turns N_2 : Number of secondary turns f : Frequency of AC supply ϕ_m : Maximum value of Flux in <u>Wb</u>
Q8	Why are transformers rated in kVA?
	▪ The copper loss (I^2R) in the transformer depends on the current 'I' through the winding. ▪ The iron or core loss depends on the voltage 'V' as the frequency of operation is constant. ▪ None of these losses depend on the power factor ($\cos \phi$) of the load. ▪ Losses decide the temperature rise & hence the rating of the transformer. ▪ As losses depend on V & I only, the transformer rating is specified as the product of voltage & current & called VA rating or KVA rating.
Q9	Enumerate the purposes which dictate the use of tertiary winding in a three winding transformer.
	• It reduces the unbalancing in the primary due to three phase unbalanced load. • It can supply additional load providing isolation from the secondary windings. • It can act as a voltage source to supply substation auxiliaries at a voltage different from those of primary and secondary winding. • Three transmission lines at different voltage levels can be interconnected by using three winding transformer. • Tertiary winding can be served for the purpose of measuring voltage of a high voltage transformer. • Static capacitors or synchronous condensers may be connected to the tertiary winding for reactive power injection into the system. • A delta-connected tertiary reduces the impedance offered to the zero sequence currents thereby allowing a large earth-fault current to flow through the protective device facilitating its proper operation. • It also permits the third harmonic current to flow.
Q10	Explain the vector groupings Yy0 and Dy1 in three phase transformers.
	Yy0 ▪ Both HV side and LV side are star connected. ▪ There is no phase shift between voltages.

		Dy1 ▪ HV side is delta connected whereas LV side is star connected. ▪ Voltage at LV side lags the voltage at HV side by 30°.
PART B		
<i>Answer any one full question from each module. Each question carries 14 marks</i>		
Module 1		
Q11	Explain the constructional details of a DC machine with the help of neat diagrams. Mention the material used for each part and give reasons for the choice of the material	
Ans	 Essential Parts are: ➤ Yoke or Frame ➤ Field System • Pole Core • Pole Shoe • Field Coils / Field Windings ➤ Armature • Armature Core • Armature Windings ➤ Commutator ➤ Brushes <u>Yoke or Frame:</u> It is the outer cover of machine. Functions: • It serves the purpose of outermost cover of the dc machine. So that machine parts get protected from moisture, dust etc. • Provide mechanical support to the poles. • It provides a path of low reluctance for the magnetic flux produced by the poles.	

	Construction material: It must be made up of some magnetic material to provide a low reluctance path for magnetic flux. • In small machines cast iron is used for the construction because it is cheap. For large machines rolled steel, cast steel etc are used. Pole Core: Every machine has even number of poles. Pole cores are bolted to the yoke. Functions: Pole core basically carries field winding which is necessary to produce flux. Construction material: Pole core is made of steel laminations Pole Shoe: Each pole core has a pole shoe of curved surface. Functions: • Pole shoes support the field windings • Spread out the flux in air gap uniformly • Increase the cross-sectional area of the magnetic circuit and thus reduce the reluctance of the flux path. Construction material: • Pole shoe is made of steel laminations Field Winding: Field windings are wound on the pole core to produce magnetic field. They are also called pole coils or exciting coils. These coils are wound over the poles in such a direction so as to form alternate north (N) and south (S) poles. Functions: • In dc machines electromagnets are used for producing necessary flux. Whenever these field windings carry current, pole cores get electro-magnetised and produce flux. Construction material: • Copper is used for the construction of field windings. Armature Core: It is cylindrical in shape and mounted on the shaft. It consists of slots on its outer periphery. Functions: • It houses armature windings on the slots provided on its periphery and causes them to rotate in the magnetic field. • It provides a path of low reluctance for the magnetic flux produced by the filed system. Construction material: As it has to provide low reluctance path for the flux, it is made of cast steel. Armature core is made of laminations to reduce eddy current loss. Armature Windings: These are placed in the slots provided on the periphery of armature core. Armature windings are of two types: Lap winding and Wave winding. Their selection will depend on the current rating of the machine. Functions:
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	• In case of generator, generation of EMF takes place in the armature windings. • In case of motors, it carries the supply current. Construction material: Armature windings are made of Copper. Commutator: The commutator is of cylindrical in shape and divided into number of segments. Number of segments is equal to the number of armature coils. Functions: • Main function of the commutator is to convert alternating current induced in the armature into unidirectional current. Construction material: Commutator is made of wedge-shaped hard drawn forged copper. Commutator segments are insulated from each other with thin layer of mica sheet. Brushes: The brushes are placed in brush holders. They rest on the commutator segments. The commutator and brush arrangement is the connecting link between the armature winding and external circuit. Functions: • Collect the current from the commutator of the machine Construction material: Brushes are made of Carbon.
Q12	a) Prepare the winding table for the simplex progressive armature winding for a 20 slot, 4 pole DC machine with two coil sides per slot. Number of parallel paths required is 4. b) An 8-pole lap connected DC armature has 960 conductors and runs at 400 rpm. The flux per pole is 40mWb. Calculate the induced emf. If the same armature is wave connected, calculate the speed at which the generator must be run to generate 400V.
Ans	a) Number of slots, $S = 20$ Number of conductors, $Z = 20 \times 2 = 40$ Number of poles, $P = 4$ Number of parallel paths = 4 = P (thus it is Lap Winding) For Lap Winding, $Y_b = \frac{Z}{P} \pm K$ Back pitch, $\frac{Z}{P} = \frac{40}{4} = 10$ Y_b should be an odd number. Thus $K=1$ $Y_b = \frac{Z}{P} + K = 10 + 1 = 11$ Since it is progressive windings, Winding pitch = 2 Front pitch, $Y_f = Y_b - Y_w = 11 - 2 = 9$

Winding Table

Back connection

Front connection

$1+11=12$

$12-9=3$

$3+11=14$

$14-9=5$

$5+11=16$

$16-9=7$

$7+11=18$

$18-9=9$

$9+11=20$

$20-9=11$

$11+11=22$

$22-9=13$

$13+11=24$

$24-9=15$

$15+11=26$

$26-9=17$

$17+11=28$

$28-9=19$

$19+11=30$

$30-9=21$

$21+11=32$

$32-9=23$

$23+11=34$

$34-9=25$

$25+11=36$

$36-9=27$

$27+11=38$

$38-9=29$

$29+11=40$

$40-9=31$

$31+11=42=2$

$42-9=33$

$33+11=44=4$

$44-9=35$

$35+11=46=6$

$46-9=37$

$37+11=48=8$

$48-9=39$

$39+11=50=10$

$10-9=1$

b)

$P=8$

$Z=960$

$\Phi=0.04 \text{ Wb}$

$N=400$

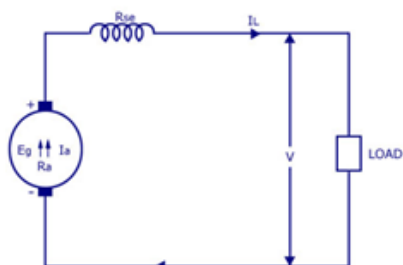
When lap connected, $A=P=8$

Induced EMF,

When wave connected and induces an EMF of 400V

	Speed required,
Module 2	
Q13	a) Derive the condition for maximum efficiency of a DC generator b) Explain any two types of DC generators with the help of neat sketches and relevant circuit equations.
Ans	a) The efficiency of DC generator is not constant but varies with load. Consider a shunt generator delivering a load current I_L at a terminal voltage V. Generator Output = VI_L Generator Input = output + losses = $VI_L + \text{variable losses} + \text{constant losses}$ = $VI_L + I_a^2 R_a + W_c$ = $VI_L + (I_L + I_{sh})^2 R_a + W_c$ The shunt field current I_a is generally small as compared to I_L and therefore can be neglected. Generator Input = $VI_L + I_L^2 R_a + W_c$ Now, Efficiency, $\eta = \text{output/input} = \frac{VI_L}{VI_L + I_L^2 R_a + W_c}$ = $\frac{1}{1 + \frac{I_L R_a}{V} + \frac{W_c}{VI_L}}$ Efficiency will be maximum when denominator is minimum. <u>ie.,</u> $\frac{d}{dI_L} (1 + \frac{I_L R_a}{V} + \frac{W_c}{VI_L}) = 0$ Or, $\frac{R_a}{V} - \frac{W_c}{VI_L^2} = 0 \Rightarrow \frac{R_a}{V} = \frac{W_c}{VI_L^2} \Rightarrow I_L^2 R_a = W_c$ <u>ie.,</u> Variable loss = Constant loss (since $I_L \approx I_a$) b)

Series Generator



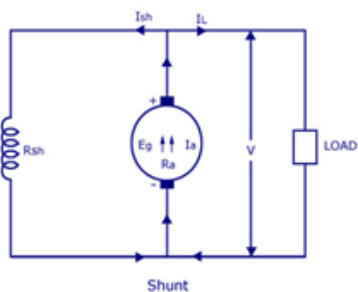
$$I_a = I_{se} = I_L$$

$$V = E_g - I_a(R_a + R_{se})$$

$$\text{Power developed, } P_g = E_g I_a$$

$$\text{Power delivered, } P_o = VI_a$$

Shunt Generator



$$I_a = I_L + I_{sh}$$

$$I_{sh} = \frac{V}{R_{sh}}$$

$$V = E_g - I_a R_a$$

$$\text{Power developed, } P_g = E_g I_a$$

$$\text{Power delivered, } P_o = VI_L$$

Q14 a) The O.C.C. of a DC generator driven at 400 rev/min is as follows:

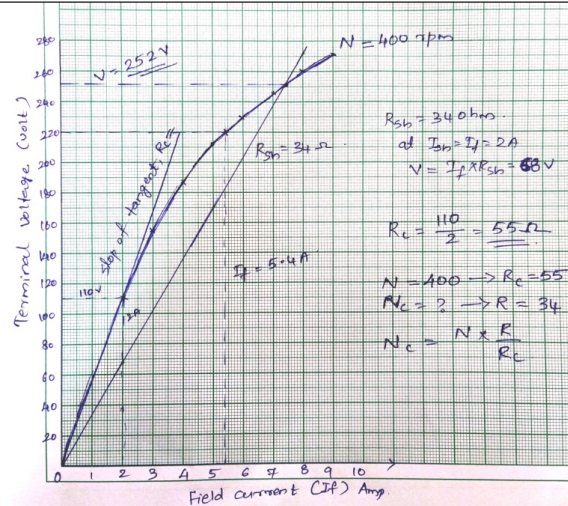
Field current (A)	2	3	4	5	6	7	8	9
Terminal voltage (V)	110	155	186	212	230	246	260	271

Find :

- the maximum voltage to which the machine will excite when run as a shunt generator at 400 rev/min with shunt field resistance equal to 34Ω .
- the additional resistance to be inserted in the shunt field circuit to reduce the O.C. voltage to 220 V.
- the critical value of the shunt field circuit resistance.
- the critical speed when the field circuit resistance is 34Ω .

b) Explain the process of commutation in DC machines?

Ans a)



a) 252 V

b) At, Terminal voltage, $V = 220 \text{ V}$

Field current. $I_f = 5.4 \text{ A}$

Thus, Field resistance, $R_{sh} = 220/5.4 = 40.74 \text{ } \Omega$

So, Additional resistance to be inserted $= 40.74 - 34 = 6.74 \text{ } \Omega$

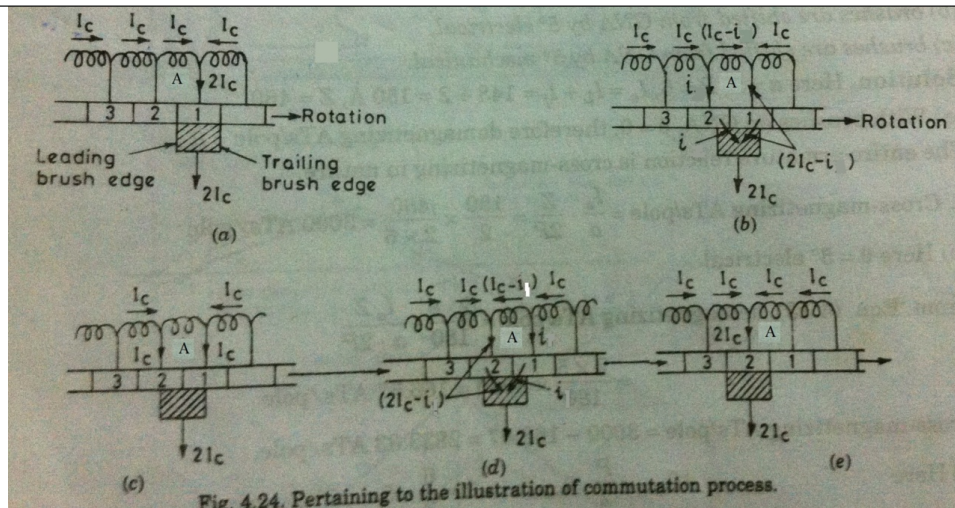
c) Critical resistance, $R_c = 55 \text{ } \Omega$

d) $N = 400 \text{ rpm} \rightarrow R_c = 55 \text{ } \Omega$

$N_c = ? \rightarrow R = 34 \text{ } \Omega$

b)

- Whenever armature conductors are alternately comes under different poles, direction of current changes.
- But in external circuit, we get unidirectional current.
- The process of converting bidirectional current in the armature conductors to unidirectional current in the external circuit by means of commutator segments and brushes is called **Commutation process**.
- In other words, the reversal of current in the armature coil by means of commutator segments and brushes is known as **Commutation**.



In Fig (a)

- Brush is fully on commutator segment 1.
- Segment 1 conducts a current of $2I_c$ to the brush.
- I_c from coil A and I_c from the adjacent coil.

In Fig (b)

- As armature rotates, brush makes contact with segment 2 and thus short circuits coil A.
- The brush is one-fourth on segment 2 and three-fourth on segment 1.
- Brush conducts a current of $2I_c$; i through segment 2 and $(2I_c - i)$ through segment 1.
- Current in coil A is reduced to $(I_c - i)$ from I_c .

In Fig (c)

- The brush is one-half on segment 1 and one-half on segment 2.
- Brush conducts a current of $2I_c$; I_c through segment 1 and I_c through segment 2.
- Current in coil A is reduced zero.

In Fig (d)

- The brush is one-fourth on segment 1 and three-fourth on segment 2.
- Brush conducts a current of $2I_c$; i through segment 1 and $(2I_c - i)$ through segment 2.
- Coil A carries a current of $(I_c - i)$ in the opposite direction.

In Fig (e)

- The brush is fully on segment 2.
- Brush conducts a current of $2I_c$; I_c through coil A and I_c from adjacent coil.
- Coil A carries a current of I_c but in the reverse direction.

- Thus coil A has undergone commutation.
- Each coil will undergo commutation in this way as it passes the brush axis.

Module 3

Q15 a)
A 4-pole, 11.19 kW, 240 V, wave connected DC shunt motor draws armature and field currents of 50 A and 1.0 A respectively while running at 1000 rpm. Its armature has 540 conductors and the armature resistance is 0.1Ω . Assuming a drop of 1 volt per brush, find (a) the torque developed (b) the shaft torque (c) useful flux / pole (d) rotational losses and (e) efficiency.

b)
Explain any two speed control methods of DC motors.

Ans

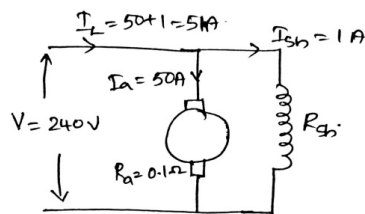
a)

a)

$$I_a = 50 \text{ A}$$

$$I_{sh} = 1 \text{ A}$$

$$\begin{aligned} I_L &= I_a + I_{sh} \\ &= 51 \text{ A} \end{aligned}$$



$$E_b = V - (I_a R_a + V_b)$$

$$V_b = 2 \text{ V (1 V / brush)}$$

$$\begin{aligned} E_b &= 240 - (51 \times 0.1 + 2) \\ &= 232.9 \text{ V} \end{aligned}$$

Power developed in the armature, $P_a = E_b I_a$.

$$\therefore \text{Torque developed, } T_e = \frac{P_a}{\omega}$$

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 1000}{60} = 104.72 \text{ rad/sec}$$

$$\text{Torque, } T_e = \frac{232.9 \times 51}{104.72} = 113.43 \text{ Nm}$$

$$\text{b) Shaft torque, } T_{sh} = \frac{P_{out}}{\omega} = \frac{11.19 \times 10^3}{104.72} = 106.86 \text{ Nm}$$

c) Torque developed,

Flux / pole,

b)

Armature Control

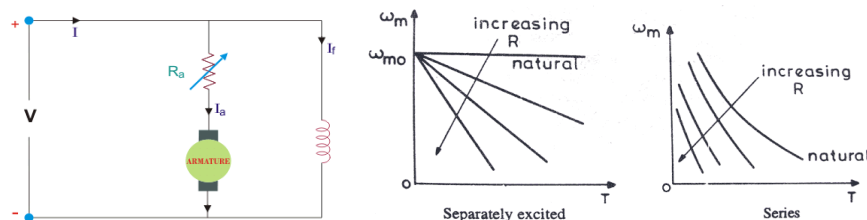
- Resistance of armature circuit is varied to control the speed
- As resistance increases, speed decreases and vice versa

Field Control

- Flux is varied to control the speed
- Flux and speed are inversely proportional
- As flux increases, speed decreases and vice versa

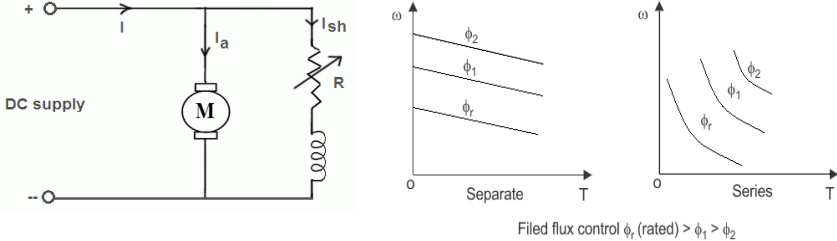
Armature Control

- ✓ This method is used when speed control below base speed is required.
- ✓ A variable resistance is directly connected in series with the armature of dc motor.
- ✓ Speed is varied by wasting power in this external resistor.
- ✓ Since it is an inefficient method of speed control, this method is not commonly used.



Field Control

- ✓ Since field flux is inversely proportional to the speed of motor, by varying field flux we can control the speed of motor.
- ✓ By decreasing the flux we can increase the motor speed and by increasing the flux we can decrease the speed.
- ✓ But, increasing of field flux is not preferred. It is because of the saturation of the magnetic core.
- ✓ So, by means of field flux control, speed of the motor is controlled only above base speed.

		
Q16	a) Derive the torque equation of a DC motor b) What is the function of a starter in a DC motor? Explain the working of a three-point starter with the help of neat diagrams	
Ans	a) Let, Z : Number of armature conductors B : Magnetic field strength in Wb/m² I : Current in armature conductor (I_a/A) Φ : Flux per pole in Wb P : Number of poles A : Number of parallel paths R_a : Radius of armature core l : Length of armature conductor Force experienced on a single armature conductor, $F = BI l \sin \theta = BI l$ Thus Torque on each conductor, $T = BI l \times R$ Torque experienced on armature, $T = BI l R \times Z$ $T = Z \frac{\phi}{a} \frac{I_a}{A} l R$ Where 'a' is the cross-sectional area of flux path per pole and $a = \frac{2\pi R l}{P}$ Therefore, Torque developed, $T_e = Z \frac{\phi}{\left(\frac{2\pi R l}{P}\right)} \frac{I_a}{A} l R = Z P \frac{\phi}{2\pi} \frac{I_a}{A}$ $T_e = 0.159 Z \phi I_a \frac{P}{A}$ $T_e = \frac{Z \phi I_a P}{2\pi A} = 0.159 Z \phi I_a \frac{P}{A}$ b)	

$$V = E_b + I_a R_a$$

$$E_b = \frac{Z \phi N}{60} \frac{P}{A} = K \phi \omega$$

During the starting of DC motor,

→ Speed is zero

→ Back emf will be zero

→ Thus

$$I_a = \frac{V}{R_a}$$

→ Since armature resistance is very low, armature current will be very large.

→ This may lead to damage of armature windings and dip in supply voltage.

This can be avoided by the use of starters.

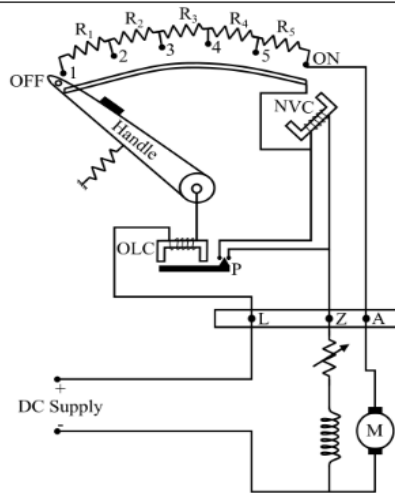
Starters introduce extra resistance in the armature path, so that armature current can be limited at safe range during starting.

THREE POINT STARTER

The circuit diagram of the three-point starter is shown in the figure. It is called three-point starter because it has three terminals viz. L, Z and A. It consists of a graded starting resistance to limit the starting current and is connected in series with the armature of the motor. The tapping points of the starting resistance are taken out to a number of studs.

The three terminals L, Z and A of the starter are connected to the positive terminal, the shunt field terminal and the armature terminal respectively. The other ends of the armature and the shunt field windings are directly connected to the negative terminal of the supply.

The no-volt trip coil (NVC) is connected in shunt field circuit, which provides protection against the open circuit in the field winding. The NVC is also known as under-voltage protection of the motor. One end of the handle is connected to the terminal L through the overload trip coil (OLC) and the other end of the handle moves against the force of control spring and makes contact with each stud during the starting period of operation. The starting resistance is cutting out gradually as the handle passes over each stud in clockwise direction.



The working of the three-point starter can be stated as follows –

Initially, the DC supply is switched on with handle in the OFF position.

The handle is now moved to the first stud. When it comes in contact with the first stud, the whole starting resistance is inserted in series with the armature winding and the shunt field winding is directly connected across the DC supply.

As the handle is gradually moved over to the final stud, the starting resistance is cut out from the armature circuit in steps. After reaching to the final stud, the handle is held magnetically by the NVC which is energized by the shunt field current.

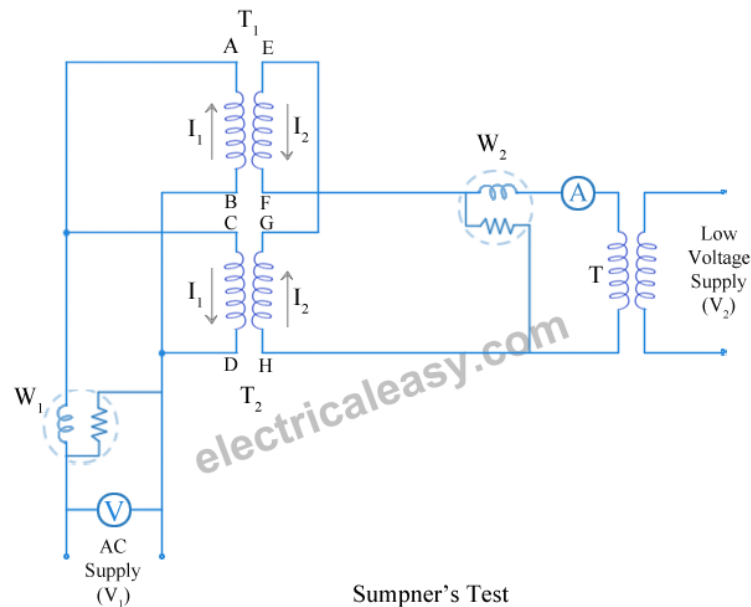
If the supply voltage is interrupted or if an open-circuit is occurred in the field circuit, the NVC is de-energized and the handle goes back to the OFF position under the pull of the control spring. If the NVC were not used, then in case of failure of supply, the handle would remain in contact with the final stud. When the supply is restored, the motor will be directly connected across the full supply voltage, resulting in an excessive armature current and may damage the motor.

If the motor is overloaded or if a short circuit is occurred, it will draw a large current from the supply. This excessive current will increase the mmf of the OLC and pull the plunger P, which short-circuits the NVC. Hence, the NVC is de-energised and the handle is pulled to the OFF position by the control spring. Therefore, the motor is automatically isolated from the supply.

Module 4

- | | |
|-----|--|
| Q17 | <p>a)
With the help of a neat diagram, explain Sumpner's test on the transformer</p> <p>b)
Explain different losses in a transformer</p> |
| Ans | <p>a)</p> |

Sumpner's test or back to back test can be employed only when two identical transformers are available. Both transformers are connected to supply such that one transformer is loaded on another. Primaries of the two identical transformers are connected in parallel across a supply. Secondaries are connected in series such that emf's of them are opposite to each other. Another low voltage supply is connected in series with secondaries to get the readings, as shown in the circuit diagram shown below.



In above diagram, T1 and T2 are identical transformers. Secondaries of them are connected in voltage opposition, i.e. EEF and EGH. Both the emf's cancel each other, as transformers are identical. In this case, as per superposition theorem, no current flows through secondary. And thus the no load test is simulated. The current drawn from V1 is $2I_0$, where I_0 is equal to no load current of each transformer. Thus input power measured by wattmeter W1 is equal to iron losses of both transformers.

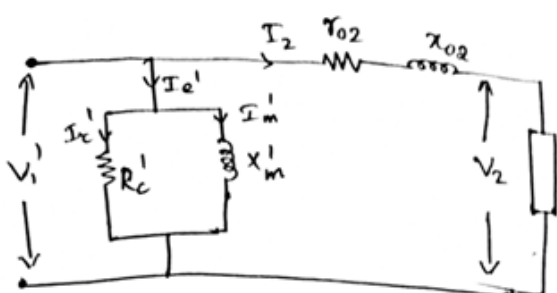
i.e. iron loss per transformer $P_i = W_1/2$.

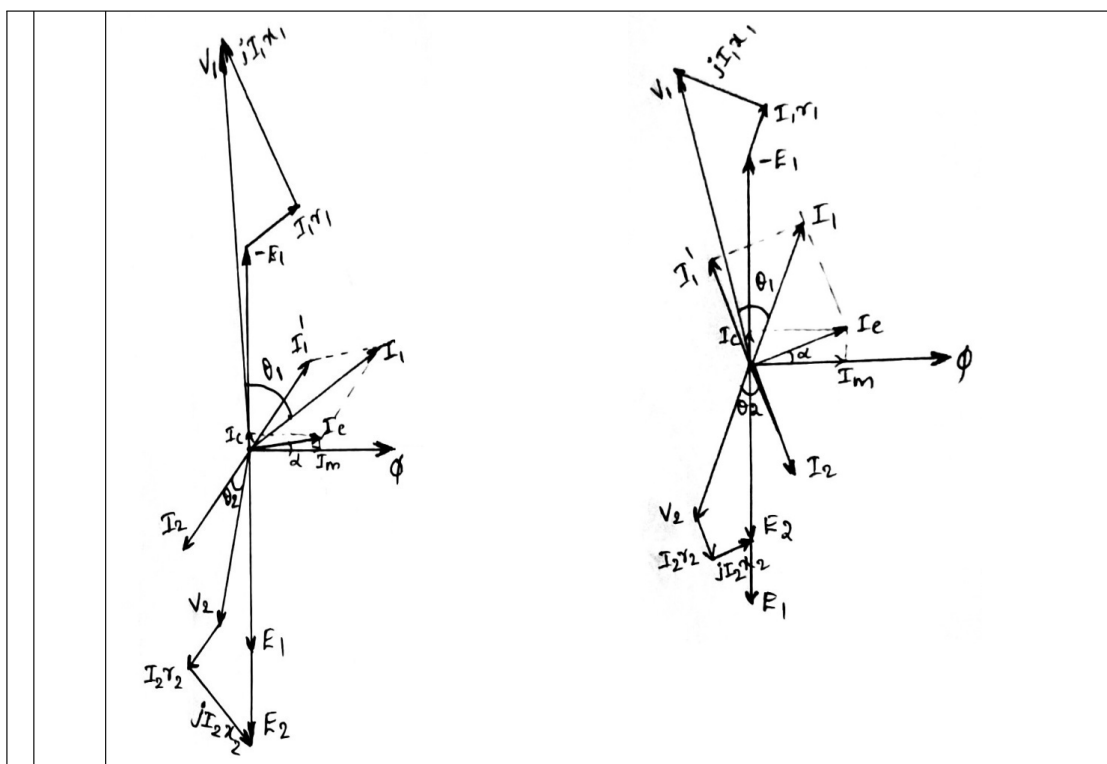
Now, a small voltage V2 is injected into secondary with the help of a low voltage transformer. The voltage V2 is adjusted so that, the rated current I_2 flows through the secondary. In this case, both primaries and secondaries carry rated current. Thus short circuit test is simulated and wattmeter W2 shows total full load copper losses of both transformers.

i.e. copper loss per transformer $P_{Cu} = W_2/2$.

From above test results, the full load efficiency of each transformer can be given as -

$$\% \text{ full load efficiency of each transformer} = \frac{\text{output}}{\text{output} + \frac{W_1}{2} + \frac{W_2}{2}} \times 100$$

	b) Losses in a transformer are Core losses (Iron losses) and Copper losses. Core losses are constant losses and it will not vary with load. Copper losses are variable losses and it varies with load. From OC test of transformer core losses can be find out where as copper losses can be determined from SC test of transformer.
Q18	a) Develop the equivalent circuit of a single phase two winding transformer referred to secondary b) Draw and explain the vector diagram of a single phase transformer supplying (i) lagging power factor load and (ii) leading power factor load
Ans	a)  $r_{02} = r_1' + r_2'$ $x_{02} = x_1' + x_2'$ $z_{02} = \sqrt{r_{02}^2 + x_{02}^2}$ $r_1' = K^2 r_1$ $x_1' = K^2 x_2$ $X_m' = K^2 X_m$ $R_c' = K^2 R_c$ b) Lagging pf load Leading pf load



Module 5

Q19	a) Draw the connection diagram of a V-V connected 3 phase transformer and discuss the operation. What is the total capacity of V-V connected transformer bank as compared with a Δ-Δ bank? b) What are the essential and desirable conditions for parallel operation of three phase transformers?
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Ans	a) V-V CONNECTION (OPEN-Δ CONNECTION) By this connection, it is possible to transform 3-phase power by means of two single-phase transformers. Here two identical transformers are needed.
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	If one transformer breaks down in a star-star connected system of three singlephase transformers, 3-phase power cannot be supplied until the defective transformer has been replaced or repaired. To eliminate this undesirable condition, in three phase system of three single-phase transformers, they are usually connected in delta-delta. In this case, one transformer breaks down, it is possible to continue supplying 3-phase power with the remaining two transformers. However, with 2 transformers, capacity of bank is reduced to 57.7% of the power transferred with 3 single-phase transformers. If one transformer is removed from delta-delta connected 3 single-phase transformers, the resulting connection is called V-V connection or Open- Δ connection. Let I_L : Line current V_L : Line voltage I_{ph} : Phase current Capacity of Δ- Δ connected transformers, $P_{\Delta-\Delta} = \sqrt{3}V_L I_L = \sqrt{3}V_L \times \sqrt{3}I_{ph} = 3V_L I_{ph}$ Capacity of Open-Δ connected transformers, $P_{V-V} = \sqrt{3}V_L I_L = \sqrt{3}V_L \times I_{ph}$ It is because, in Open-Δ connection phase current is same as line current. $\frac{P_{V-V}}{P_{\Delta-\Delta}} = \frac{3V_L I_{ph}}{\sqrt{3}V_L I_{ph}} = \frac{1}{\sqrt{3}} = 0.577 \text{ or } 57.7\%$ b) • The line voltage ratio of the transformers must be same. • The transformers should have equal per unit impedance. • The transformers should have the same polarity. • There should not be any phase difference between the secondary line voltages of all the transformers. • The phase sequence of secondary line voltages of all the transformers should be same.
Q20	a) Derive an expression for the saving in copper effected by using an autotransformer in place of a two winding transformer b) A 3φ step down transformer is connected to 3.3 kV supply mains and takes 30A. Calculate its secondary line voltage and line current for the following connections if the ratio of turns per phase is 14 (i) Y-Y (ii) Y- Δ (iii) Δ – Y (iv) Δ – Δ.
Ans	a) For any winding, cross-section of winding is proportional to the current flowing through it and length of winding is proportional to the number of turns.

Hence, weight of copper needed for winding is proportional to the product of current and number of turns.

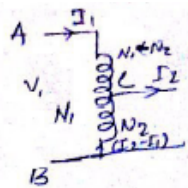
In 2-winding transformer,

Weight of copper of primary winding $\propto N_1 I_1$

Weight of copper of secondary winding $\propto N_2 I_2$

Total weight of copper of 2-winding transformer $\propto N_1 I_1 + N_2 I_2$

In case of autotransformer,



Weight of copper of section AC $\propto (N_1 - N_2) I_1$

Weight of copper of section BC $\propto N_2 (I_2 - I_1)$

Total weight of copper of 2-winding transformer $\propto (N_1 - N_2) I_1 + N_2 (I_2 - I_1)$

$$\therefore \frac{W_{TW}}{W_{AT}} = \frac{N_1 I_1 + N_2 I_2}{N_1 I_1 - N_2 I_1 + N_2 I_2 - N_2 I_1}$$

$$= \frac{N_1 I_1 + N_2 I_2}{N_1 I_1 + N_2 I_2 - 2 N_2 I_1}$$

$$K = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

$$\therefore \frac{W_{TW}}{W_{AT}} = \frac{N_1 I_1 + K N_1 \frac{I_1}{K}}{N_1 I_1 + K N_1 \frac{I_1}{K} - 2 \times K N_1 I_1}$$

$$= \frac{2 N_1 I_1}{2 N_1 I_1 - 2 K N_1 I_1} = \frac{1}{1 - K}$$

$$W_{AT} = (1-k) W_{TW}$$

$$\therefore \text{Savng of Cu} = W_{TW} - W_{TW}(1-k) \\ = \underline{\underline{k \times W_{TW}}} \quad (\text{for stepdown xmer})$$

b)

Given,

$$V_{L1} = 3300 \text{ V}$$

$$I_{L1} = 30 \text{ A}$$

$$N_1 / N_2 = 14$$

For Y-Y connection

$$\text{Phase voltage at primary side, } V_1 = V_{L1} / \sqrt{3} = 1905.26 \text{ V}$$

$$\text{Phase current at primary side, } I_1 = I_{L1} = 30 \text{ A}$$

We have,

Phase voltage at secondary side,

Phase current at secondary side,

Thus,

$$V_{L2} = 235.71 \text{ V}$$

$$I_{L2} = 420 \text{ A}$$

For Y-Δ connection

$$\text{Phase voltage at primary side, } V_1 = V_{L1} / \sqrt{3} = 1905.26 \text{ V}$$

$$\text{Phase current at primary side, } I_1 = I_{L1} = 30 \text{ A}$$

We have,

Phase voltage at secondary side,

Phase current at secondary side,

	Thus, $V_{L2} = 136.09 \text{ V}$ $I_{L2} = 727.46 \text{ A}$ For Δ-Y connection Phase voltage at primary side, $V_1 = V_{L1} = 3300 \text{ V}$ Phase current at primary side, $I_1 = I_{L1} / \sqrt{3} = 17.32 \text{ A}$ We have, Phase voltage at secondary side, Phase current at secondary side, Thus, $V_{L2} = 235.71 * \sqrt{3} = 408.26 \text{ V}$ $I_{L2} = 242.48 \text{ A}$ For Δ-Δ connection Phase voltage at primary side, $V_1 = V_{L1} = 3300 \text{ V}$ Phase current at primary side, $I_1 = I_{L1} / \sqrt{3} = 17.32 \text{ A}$ We have, Phase voltage at secondary side, Phase current at secondary side, Thus, $V_{L2} = 235.71 \text{ V}$ $I_{L2} = 242.48 * \sqrt{3} = 420 \text{ A}$
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